

# Task 14: Generic Programming in Java

This report explains the implementation of Generic Programming in Java using generic classes, generic methods, bounded type parameters, and wildcards.

---

## Objectives

- Understand Java Generics
- Create reusable and type-safe code
- Implement Generic Classes and Methods
- Use Bounded Type Parameters and Wildcards

## Tools Used

- Java IDE (IntelliJ IDEA / Eclipse / VS Code)

## Java Code

```
class Box<T> {
    private T value;

    public void setValue(T value) {
        this.value = value;
    }

    public T getValue() {
        return value;
    }

    public void display() {
        System.out.println("Stored Value: " + value);
    }
}

class GenericMethod {
    public static <T> void printData(T data) {
        System.out.println("Data: " + data);
    }
}

class Calculator<T extends Number> {
    public double square(T num) {
        return num.doubleValue() * num.doubleValue();
    }
}

class WildcardDemo {
    public static void showNumbers(Box<? extends Number> box) {
        System.out.println("Wildcard Value: " + box.getValue());
    }
}

public class Main {
    public static void main(String[] args) {
        Box<String> stringBox = new Box<>();
        stringBox.setValue("Hello Java Generics");
        stringBox.display();

        Box<Integer> intBox = new Box<>();
    }
}
```

```

        intBox.setValue(100);
        intBox.display();

        GenericMethod.printData("Java");
        GenericMethod.printData(500);

        Calculator<Integer> calc = new Calculator<>();
        System.out.println("Square: " + calc.square(5));

        WildcardDemo.showNumbers(intBox);
    }
}

```

## Program Output

```

Stored Value: Hello Java Generics
Stored Value: 100
Data: Java
Data: 500
Square: 25.0
Wildcard Value: 100

```

## Explanation

**Generic Class:** A generic class can work with different data types.

**Generic Method:** A method that accepts any type of parameter.

**Bounded Type Parameter:** Restricts the type to a specific class such as Number.

**Wildcards:** Wildcards (?) allow flexibility while handling generic objects.

## Time Complexity

| Operation          | Complexity |
|--------------------|------------|
| Set Value          | O(1)       |
| Get Value          | O(1)       |
| Print Data         | O(1)       |
| Square Calculation | O(1)       |

## Interview Questions

### Q1: Why use Generics?

Generics provide type safety, code reusability, and reduce runtime errors.

### Q2: What is Type Erasure?

Type Erasure means Java removes generic type information during compilation.

## Conclusion

This project demonstrates the use of Generic Programming in Java. It helps developers create reusable, flexible, and type-safe components for modern Java applications.